Vishal Gupta

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CONTACT

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Google Scholar

Marshall School of Business
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RESEARCH INTERESTS

Data-driven optimization in settings with scarce or low-precision data. Decision-Focused Learning. Applications in artificial intelligence, causal inference and supply-chain.

EMPLOYMENT

Marshall School of Business, Los Angeles, CA **2014 – Pres.**
- Dean's Associate Professorship in Business Administration 2024 – Pres.
- Associate Professor (by courtesy) of Industrial & Systems Engineering 2022 – Pres.
- Affiliate Faculty, Center for Artificial Intelligence and Society 2019 – Pres.
- Associate Professor (w. tenure) of Data Sciences and Operations 2021 – 2024
- Assistant Professor of Data Sciences and Operations 2014 – 2021

The Wharton School at the University of Pennsylvania **Spring 2023**
- Visiting Scholar

National University of Singapore, Singapore **Fall 2022**
- Visiting Professor, Institute for Operations Research and Analytics
- Visiting Professor, NUS Business School, Analytics and Operations

Analytics Operations Engineering, Inc., Boston, MA **Summer 2011**
- Summer Associate

Barclays Capital, New York, NY **2005 – 2009**
- New York Head of Commodities Tactical Modeling 2008 – 2009
- Manager, Quantitative Analytics Commodities Modeling Group 2007 – 2008
- Analyst, Quantitative Analytics Commodities Modeling Group 2005 – 2007

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA **2009 – 2014**
- Ph.D. in Operations Research
  - Thesis: Data-Driven Models for Uncertainty and Behavior
  - Advisor: Prof. Dimitris Bertsimas

University of Cambridge, Cambridge, England **2004 – 2005**
- Part III Mathematics Tripos — Graduated with "Distinction"
  - Essay: Hedging Financial Derivatives as a Differential Game

Yale University, New Haven, CT **2000 – 2004**
- B.A. in Mathematics and Philosophy
- Graduated with Honors, Magna Cum Laude; Phi Beta Kappa

HONORS AND AWARDS

Global Supply Chain Excellence Summit: Faculty Excellence Award **2024**
Awarded to one faculty by the Kendrick Supply Chain Institute for achievements across teaching and research of supply-chain analytics.

Evan C. Thompson Award for Faculty Mentoring and Leadership **2024**
Awarded for excellence in mentoring of junior faculty, graduate students, and undergraduates.

Marshall Leadership Program **2024**
Selected for professional development program focused on preparing promising faculty and staff for leadership.

Dean's Award for Community **2022**
Awarded to one faculty member each year at USC Marshall who has "risen above the call of duty to make Marshall a better place."

Daniel H. Wagner Prize for Excellence in the Practice of Advanced Analytics and Operations Research **2021**
Awarded for paper "Interpretable OR for High-Stakes Decision-Making: Designing the Greek COVID-19 Testing System" (with H. Bastani and K. Drakopoulos). The Wagner Prize recognizes strong mathematics applied to practical problems and is awarded to a paper demonstrating strong analytical content, good writing, and verifiable success in practice.

Pierskalla Best Paper Competition **2021**
Awarded by the Health Applications Society of INFORMS for the paper "Deploying an Artificial Intelligence System for COVID-19 Testing at the Greek Border" (with H. Bastani, K. Drakopoulos, J. Vlachogiannis, C. Hadjicristodoulou, P. Lagiou, G. Magiorkinis, D. Paraskevis, S. Tsiodras). INFORMS selects 3–5 finalists each year to recognize research excellence in the field of health care management science.

Public Sector Operations Research Best Paper Award, 2nd Place **2021**
Awarded by the Public Sector Operations Research Society of INFORMS for the paper "Deploying an Artificial Intelligence System for COVID-19 Testing at the Greek Border" (with H. Bastani et al.). INFORMS selects 4 papers judged to be the best quality across all application areas represented within public sector operations research.

Post-Pandemic Supply-Chain and Healthcare Management Best Paper Competition Finalist **2021**
Awarded for the paper "Deploying an Artificial Intelligence System for COVID-19 Testing at the Greek Border." One of 6 finalists.

Dr. Jagdish Sheth Impact of Research on Practice Award **2021**
Awarded to a USC Marshall faculty each year whose research has meaningfully changed the practice of business, regulators, or society at large. Awarded in 2021 for ongoing collaboration with the Greek Government in their management of the COVID-19 pandemic.

Dean's Award for Research Excellence **2020**
Awarded to a faculty member at USC Marshall whose research meets the highest aspirations of the Marshall School and USC.

Management Science Meritorious Service Award **2018, 2019, 2020**
Awarded to a select group of reviewers for consistently writing "timely, unbiased, and thoughtful" referee reports.

Pierskalla Best Paper Competition Finalist **2018**
Awarded by the Health Applications Society of INFORMS for the paper "Maximizing Intervention Effectiveness" (with B.R. Han, S.H. Kim and H. Paek).

Service Science Best Paper Competition Finalist **2018**
Awarded by the INFORMS Service Science Section for the paper "Value of Personalized Pricing" (with A. Elmachtoub and M. Hamilton).

POMS CHOM Best Paper Competition Finalist **2018**
Awarded by the College of Healthcare Operations Management (CHOM) for the paper "Max-

imizing Intervention Effectiveness" (with B.R. Han, S.H. Kim and H. Paek).

Evan C. Thompson Teaching and Learning Innovation Award **2016**
Awarded for curriculum redesign of BUAD 425: Data-Analysis for Decision-Making.

George Nicholson Student Paper Competition Finalist **2013**
Awarded for the paper "Data-Driven Robust Optimization" (with D. Bertsimas and N. Kallus).

Best Student Paper Prize, MIT Operations Research Center **2013**
Awarded for the paper "Robust SAA" (with D. Bertsimas and N. Kallus).

MIT Teaching Certificate **2013**

Nominated for Excellence in Teaching Award **2012**
For teaching assistant role in the MBA Core Course "Data, Models and Decisions" at MIT Sloan. Selected by MBA students.

Charles M. Vest Presidential Fellowship for Doctoral Studies **2009 – 2010**
Awarded to three first-year graduate students at MIT across all fields to support their doctoral work.

Paul Mellon Fellowship for Graduate Research **2005**
Awarded to one graduating Yale senior to fully support two years of study at the University of Cambridge, UK.

Timothy Dwight Masters Cup **2004**
Awarded each year to a graduating senior who exemplifies high academic rank, scholarly achievement, and the values of Timothy Dwight College at Yale University.

Grants

NSF: AI Institute for Advances in Optimization **2026 – 2031**
(NSF 23-610). Role: Co-PI (USC Site). $5,600,000

Research Catalyst: Tackling Data Scarcity in Supply-Chains through Decision-Focused Data Aggregation **2026 – 2027**
Role: Sole Principal Investigator. ***(pending)*** *$67,617.08*

USC Undergraduate Research Associates Program **2026 – 2027**
How Noisy is Too Noisy? Revisiting Benchmarks in Decision-Focused Learning. $6,600

Enabling Trustworthy Communication in Supply-Chain through Blockchain **2025**
(USC Marshall Business of Blockchain Initiative). Role: Co-PI. Awarded to fund a one-year postdoctoral appointment; unfilled following subsequent University budget reductions. ***(funding withdrawn)*** *1 postdoc yr*

NRT-AI: Integrating Artificial Intelligence and Operations Research Technologies **2024 – 2029**
(NSF FAIN: 2940542). Role: Co-PI. $2,940,542

Optimization in the Small Data Regime **2017 – 2021**
(NSF CMMI-1661732). Role: Sole Principal Investigator. $221,592

Small Data Linear Optimization **2017 – 2025**
(Outlier Research Grant, iORB, USC). Role: Principal Investigator. $25,000

Publications

Asterisk (*) indicates a student or post-doc co-author. All authorship is alphabetical unless otherwise indicated.

1. "Modeling First: Rethinking Undergraduate Operations Management with AI," with A. Daw and P. Rusmevichientong. **INFORMS Transactions on Education, 2026** (Special Issue on Generative AI in Teaching).

2. "Decision-Focused Learning with Directional Gradients," with M. Huang. **NeurIPS 2024**. Extended journal version in preparation.

3. "Beyond Discretization: Learning the Optimal Solution Path," with Q. Dong* and P. Grigas. **AISTATS 2025**.

4. "Decision-Aware Denoising," with M. Huang* and P. Rusmevichientong. **Management Science, Forthcoming**.

5. "Balanced Off-Policy Evaluation for Personalized Pricing," with A. Elmachtoub and Y. Zhao*. **AISTATS 2023**.

6. "Debiasing In-Sample Policy Performance for Small-Data, Large-Scale Optimization," with M. Huang* and P. Rusmevichientong. **Operations Research, 2024**.

7. "Interpretable Operations Research for High-Stakes Decisions: Designing the Greek COVID-19 Testing System," with H. Bastani, K. Drakopoulos, J. Vlachogiannis, C. Hadjicristodoulou, P. Lagiou, G. Magiorkinis, D. Paraskevis, S. Tsiodras. **INFORMS Journal on Applied Analytics, 2022**.
   - Winner of the 2021 Wagner Prize for Excellence in the Practice of Advanced Analytics and Operations Research.

   Note: Authors H. Bastani, K. Drakopoulos and V. Gupta contributed equally to the work.

8. "Efficient and Targeted COVID-19 Border Testing via Reinforcement Learning," with H. Bastani, K. Drakopoulos, J. Vlachogiannis, C. Hadjicristodoulou, P. Lagiou, G. Magiorkinis, D. Paraskevis, S. Tsiodras. **Nature, 2021**.
   - Winner of the 2021 Pierskalla Best Paper Competition.
   - 2nd Place in the 2021 Public Sector Operations Research Best Paper Competition.
   - Finalist in the 2021 Post-Pandemic Supply-Chain and Healthcare Management Best Paper Competition.
   - Selected for Spotlight Presentation at the Reinforcement Learning for Real-Life Workshop (ICML 2021).

   Note: Authors H. Bastani, K. Drakopoulos and V. Gupta contributed equally to the work.

9. "Data-Pooling in Stochastic Optimization," with N. Kallus. **Management Science, 2022**.

10. "Value of Personalized Pricing," with A. Elmachtoub and M. Hamilton*. **Management Science, 2021**.
    - Finalist in the 2018 INFORMS Service Science Best Paper Competition.
    - Accepted to 15th Conference on Web and Internet Economics (WINE), 2019.

11. "Small-Data, Large-Scale Linear Optimization with Uncertain Objectives," with P. Rusmevichientong. **Management Science, 2021**.

12. "Maximizing Intervention Effectiveness," with B.R. Han*, S.H. Kim, and H. Paek. **Man-**

agement Science, 2020.
- Finalist in the 2018 Pierskalla Best Paper Competition.
- Finalist in the 2018 POMS College of Healthcare Operations (CHOM) Best Paper Competition.

13. "Near-Optimal Bayesian Ambiguity Sets for Distributionally Robust Optimization" (Single Author Work). **Management Science, 2019**.

14. "Robust Sample Average Approximation," with D. Bertsimas and N. Kallus. **Mathematical Programming, 2018**.
    - Awarded 2013 Best Student Paper, MIT Operations Research Center.

15. "Data-Driven Robust Optimization," with D. Bertsimas and N. Kallus. **Mathematical Programming, 2018**.
    - Finalist in the 2013 George Nicholson Student Paper Competition.

16. "A Comparison of Monte Carlo Tree Search and Rolling Horizon Optimization for Large-Scale Dynamic Resource Allocation Problems," with D. Bertsimas, D. Griffith, M. Kochenderfer, and V. Mišić. **European Journal of Operational Research, 2017**.

17. "A Course on Advanced Software Tools for Operations Research and Analytics," with I. Dunning, A. King, J. Kung, M. Lubin and J. Silberholz. **INFORMS Transactions on Education, 2015**.

18. "Data-Driven Estimation in Equilibrium using Inverse Optimization," with D. Bertsimas and I. Ch. Paschalidis. **Mathematical Programming, 2015**.

19. "Inverse Optimization: A New Perspective on the Black-Litterman Model," with D. Bertsimas and I. Ch. Paschalidis. **Operations Research, 2012**.

Under Review

20. "3D-Printing for Supply Chain Resilience," with Z. He* and N. Vyas.

Workshop Papers

Peer-reviewed, non-archival.

1. "Hindsight Is 20/20: Making Regret Meaningful in Decision-Focused Learning," with S. Ahmed*. MLxOR Workshop, NeurIPS 2026.

2. "Differentiating Through Dual Prices: End-to-End Policy Learning Under Capacity Constraints," M. Haghi*, N. Kelidari*, M. Salmani*, V. Gupta, and P. Vayanos. MLxOR Workshop, NeurIPS 2026.
   Note: Author order is not alphabetical.

Invited Book Chapters

"Chatocrates: An AI-Powered, Socratic Chatbot for Teaching AI Ethics," chapter in "Embracing AI in Business Education: A Case Study of USC Marshall School of Business" (with R. Randhawa, P. Cardon, et al.), **SSRN white paper, 2024**.
- Note: This is a comprehensive Case Study with many authors. My contribution resembles a "book chapter" in the larger work.

"Optimization in the Small-Data, Large-Scale Regime" (Single Author Work). **Joint Learning and Optimization in Operations Management, 2022**. Editors: Xi Chen, Stefanus Jasin, and Cong Shi.

"Reinforcement Learning for Public Health: Targeted COVID-19 Screening," with H. Bastani and K. Drakopoulos. **Artificial Intelligence for Social Impact, 2022**. Editors: Fei Fang,

Bryan Wilder and Milind Tambe.

INVITED TALKS

"Contextual Optimization" (Co-Instructor)
- AI-Score Summer School, Chicago, IL. (Apr. 2026)

"A Unifying Perspective of Surrogate Losses for Linear Decision-Focused Learning"
- Operations Management Departmental Seminar, Booth School of Business, University of Chicago, IL. (Apr. 2026)
- Artificial Intelligence Seminar, USC Information Sciences Institute, Los Angeles, CA. (Apr. 2026)

Keynote: "Data and Dynamic Rate Inference"
- FreightTech 2025, Global Freight Digitalization Conference, Barcelona, Spain. (Sep. 2025)

"Surrogates for Decision-Aware Learning"
- Departmental Seminar, UNC Kenan-Flagler Business School. (Dec. 2024)

"Decision-Aware Denoising"
- 2024 INFORMS Annual Meeting, Seattle, WA. (Oct. 2024)

Panelist: "AI in Higher Education"
- USC Marshall 12th Annual Global Supply Chain Summit.

"Learning Best-in-Class Policies in the Predict-Then-Optimize Framework"
- 25th International Symposium on Mathematical Programming (ISMP), Vancouver, Canada. (May 2024)

"AI: Pitfalls, Perils and Promise"
- The New Analytics MBA, Columbia Business School, New York. (2023)

Invited Discussant / Lead Researcher, CCC-INFORMS-ACM SIGAI Workshop on Artificial Intelligence and Operations Research
- Session on "Deploying ML Models in Low-Data Regimes." (Mar. 2024)

"Decision-Aware Data Aggregation"
- 2023 INFORMS Annual Meeting, Phoenix, AZ. (Oct. 2023)
- Departmental Seminar, Goizueta Business School. (Sep. 2023)
- SIAM Conference on Optimization, Seattle, WA. (2023)

"Debiasing In-Sample Policy Performance"
- Departmental Seminar, UT Austin McCombs School of Business. (Oct. 2023)
- Departmental Seminar, UBC Sauder School of Business. (Sep. 2023)
- Departmental Seminar, The Wharton School, University of Pennsylvania. (Apr. 2023)
- Departmental Seminar, Duke Fuqua. (Apr. 2023)

"Contextual Stochastic Optimization with Panel Data"
- Departmental Seminar, Singapore Management University. (Oct. 2022)
- Departmental Seminar, Singapore University of Technology and Design. (Oct. 2022)

Keynote: "The Small-Data, Large-Scale Optimization Regime: The Future of Analytics"
- Analytics for X Conference, Institute for Operations Research and Analytics, National University of Singapore. (Oct. 2022)

"Interpretable Operations Research for High Stakes Decision-Making: Designing the Greek COVID-19 Testing System"
- Indian Institute of Management Ahmedabad (IIM), Research and Publication Webinar Series. (Apr. 2022)
- 2022 INFORMS Business Analytics Conference, Wagner Reprise Session, Houston, TX. (Apr. 2022)
- 2021 INFORMS Annual Meeting, Wagner Prize Session, Anaheim, CA. (Oct. 2021)

"Real-Time, Targeted COVID-19 Screening at the Greek Border"
- INFORMS Health Application Society, Online Seminar. (June 2022)
- Guest Lecture, "Analytics for Social Impact," USC Viterbi School of Engineering, Los Angeles, CA. (Apr. 2022)
- INFORMS Annual Meeting, Anaheim, CA. (Oct. 2021)
- INFORMS Annual Meeting, Virtual. (Nov. 2020)
- Dean's Dialogue Webinar Series, USC Marshall. (Nov. 2020)
- Guest Lecture, "IEOR 4650: Business Analytics," Columbia University, New York, NY. (Nov. 2020)
- Operations Management Seminar, Indian School of Business, Virtual. (Oct. 2020)

"Decision-Making under Data Scarcity"
- Assistant Professor Research Day, USC Marshall, Los Angeles, CA. (Dec. 2020)
- Operations Management Seminar, USC Marshall, Los Angeles, CA. (Oct. 2020)

"Data-Pooling in Stochastic Optimization"
- Guest Lecture, "OIDD 941," The Wharton School, Philadelphia, PA. (Apr. 2023)
- Technology and Operations Departmental Seminar, University of Michigan Ross School of Business, Ann Arbor, MI. (Dec. 2019)
- Guest Lecture, "IEOR 8100 Prescriptive Analytics," Columbia University, New York, NY. (Nov. 2019)
- Decisions, Operations and Technology Management Seminar, UCLA Anderson School of Management, Los Angeles, CA. (Nov. 2019)
- Desautels Faculty of Management Departmental Seminar, McGill University, Montreal, Canada. (Oct. 2019)
- INFORMS Annual Meeting, Seattle, WA. (Oct. 2019)
- Industrial and Systems Engineering Departmental Seminar, University of Southern California, Los Angeles, CA. (Sep. 2019)
- DSO Graduate Research Forum, USC Marshall, Los Angeles, CA. (Sep. 2019)
- Operations Management Departmental Seminar, Booth School of Business, University of Chicago, IL. (Sep. 2019)
- 6th International Conference on Continuous Optimization (ICCOPT), Berlin, Germany. (Aug. 2019)
- Operations and Information Technology Departmental Seminar, Stanford Graduate School of Business, Palo Alto, CA. (May 2019)
- Southern California OR/OM Day, UC Irvine, Irvine, CA. (May 2019)
- Models and Algorithms for Sequential Decision-Making Problems Under Uncertainty Workshop, Banff International Research Station, Canada.
- Joint IEOR and DRO Departmental Seminar, Columbia University, New York, NY.

- INFORMS Annual Meeting, Phoenix, AZ. (Nov. 2018)

"Probability Guarantees in Data-Driven Robust Optimization"
- Guest Lecture, ISyE Reading Group, USC Viterbi, Los Angeles, CA. (Oct. 2019)

"Operations Research and Analytics Education" (Panel Speaker)
- 65th Operations Research Center Reunion, MIT, Cambridge, MA. (Nov. 2018)

"Optimization in the Small-Data, Large-Scale Regime"
- Management Sciences and Operations Department Seminar, Imperial College School of Business, London, UK. (July 2018)
- 29th European Conference on Operations Research (EURO), Valencia, Spain. (July 2018)
- 23rd International Symposium on Mathematical Programming (ISMP), Bordeaux, France. (July 2018)
- Decision Sciences Group, Duke Fuqua School of Business, Durham, NC. (May 2018)
- Technology and Operations Management Group Seminar, INSEAD, Paris, France. (Apr. 2018)
- Operations and Logistics Division Seminar, UBC Sauder School of Business, Vancouver, Canada. (Jan. 2018)
- DSO Graduate Research Forum, USC Marshall, Los Angeles, CA. (Dec. 2017)
- INFORMS Annual Meeting, Houston, TX. (Oct. 2017)
- INFORMS Annual Meeting, Nashville, TN. (Nov. 2016)
- 5th International Conference on Continuous Optimization (ICCOPT), Tokyo, Japan. (Aug. 2016)

"Calibrating Uncertainty Sets in the Small-Data, Large-Scale Regime"
- Distributionally Robust Optimization Workshop, Banff International Research Station, Canada. (Mar. 2018)

"Maximizing Intervention Effectiveness"
- International Conference on Stochastic Optimization (ICSP), Trondheim, Norway. (July 2019)

"Data-Driven Distributionally Robust Optimization"
- Electrical Engineering Group, USC Viterbi, Los Angeles, CA. (Jan. 2016)

"Near-Optimal Bayesian Ambiguity Sets in Distributionally Robust Optimization"
- INFORMS Annual Meeting, Philadelphia, PA. (Nov. 2015)
- 22nd International Symposium on Mathematical Programming (ISMP), Pittsburgh, PA. (July 2015)
- British-French-German (BFG) Conference on Optimization, London, UK. (June 2015)
- Southern California OM/OR Conference, UCLA. (May 2015)
- INFORMS Annual Meeting, San Francisco, CA. (Nov. 2014)

"Modeling Uncertainty in Optimization"
- DSO Graduate Research Forum, USC Marshall, Los Angeles, CA. (Feb. 2015)

"Data-Driven Robust Optimization" (Job Market Talk)
- Carnegie Mellon University, Pittsburgh, PA. (Feb. 2014)
- Industrial and Operations Engineering, University of Michigan, Ann Arbor, MI. (Feb. 2014)

- McCombs Business School, University of Texas at Austin, TX. (Feb. 2014)
- USC Marshall School of Business, Los Angeles, CA. (Feb. 2014)
- NYU Stern School of Business, New York, NY. (Jan. 2014)
- London Business School (LBS), London, UK. (Jan. 2014)
- Operations Management Seminar, MIT Sloan School of Management, Cambridge, MA. (Nov. 2013)
- INFORMS Annual Meeting, Minneapolis, MN. (Oct. 2013)
- MSOM Conference, Paris, France. (July 2013)
- Conference on Computational Management Science (CMS), Montreal, Canada. (May 2013)

"Inverse Optimization Approaches to Estimation"

- INFORMS Annual Meeting, Phoenix, AZ. (Oct. 2012)
- 21st International Symposium on Mathematical Programming (ISMP), Berlin, Germany. (June 2012)

"Constructing Investor Risk Preferences from Data"

- INFORMS Annual Meeting, Minneapolis, MN. (Oct. 2013)
- INFORMS Annual Meeting, Phoenix, AZ. (Oct. 2012)

"Inverse Optimization: A New Perspective on the Black-Litterman Model"

- INFORMS Annual Meeting, Charlotte, NC. (Nov. 2011)

Teaching

AI-Score Summer School: Contextual Optimization **2026**
PhD Students. Co-Instructor, Course Development.
AI-Score Summer School is a joint effort from the CS and OR communities supported by the Computing Research Association's Computing Community Consortium (CCC), INFORMS and SIGAI. I was asked to co-develop a course for the summer school on decision-focused learning / contextual optimization.

DSO/CSCI/ISE 619: End-to-End Learning and Optimization **2026**
PhD and Graduate Students, USC Marshall / USC Viterbi. Instructor, Course Development.
New core class in the Operations Research and Artificial Intelligence certificate program, targeting both DSO and USC Viterbi (computer science and ISE) PhD students. Covers foundational results in decision-focused learning, with an emphasis on theory, modeling, and implementation. Fully designed and taught course.

BUAD 313: Advanced Operations Management **2025**
Undergraduate Core, USC Marshall. Instructor.
Advanced-track core class focused on mathematical modeling. Spans optimization modeling (linear, mixed-binary) and simulation modeling (discrete-event simulation and queuing via Python). Strongly integrates AI training through pair-programming assistants.

DSO 699: Stochastic Modeling for Optimization and Learning **2021, 2024**
PhD and Graduate Students, USC Marshall. Instructor, Course Development.
New core class in the Operations Management PhD Program. Covers fundamental results in probability theory, concentration of measure, and convergence theorems with applications in data-driven operations management and prescriptive analytics.

BUAD 498: AI: Seed for Change or Existential Threat? **2022, 2024**
Undergraduate Elective, USC Marshall. Instructor, Course Development.
Part of Marshall's "Innovative Courses for Undergraduates" program. Exposes students to

both the promise and perils of AI, centered on guest speakers from industry, non-profit, and government.

PhD Mini-Course: Foundations of Decision-Aware Learning **2022**
PhD and Graduate Students, National University of Singapore / Institute for Operations Research and Analytics. Instructor, Course Development.
Introduces graduate students to the principal theoretical tools used when designing and analyzing decision-aware algorithms for data-driven optimization.

BUAD 493/494: Marshall Honors Research Seminar **2021**
Undergraduate, USC Marshall. Instructor, Course Development.
Two-semester seminar to advise business honors students on conducting independent research in data science and operations, culminating in their undergraduate thesis.

DSO 699: Workshop on Expositional Writing in Mathematics **2021**
PhD and Graduate Students, USC Marshall. Instructor, Course Development.
New core class in the Operations Management PhD Program. Helps students focus on best practices when writing mathematics, particularly for peer-reviewed journals.

DSO 670: Data-Driven Optimization: Theory, Methods and Current Themes **2020**
PhD and Graduate Students, USC Marshall. Instructor.
A PhD seminar course covering recent papers.

BUAD 311: Introduction to Operations Management **2015, 2019 – 2021**
Undergraduate Core, USC Marshall. Instructor.

BUAD 425: Data-Analysis for Decision Making **2016, 2017**
Undergraduate Core, USC Marshall. Instructor, Course Coordinator.
Redesigned course with new emphasis on critical thinking and decision-making. Authored cases, created online videos, and developed new curriculum content.

15.S60: Software Tools for Operations Research **2013, 2014**
PhD, MBA and Executive MBA Elective, MIT Sloan School of Management. Instructor.
Designed new course with primary role in curriculum development. Oversaw course logistics and lectured on select topics in convex optimization.

Mentorship

PhD Mentorship — Advisor:

- Minghua Zhang, USC Marshall (2025 – Pres.)
- Luyang Zhang, USC Marshall (2022 – 2024). Transferred to IS Program at Carnegie Mellon University.
- Michael Huang, USC Marshall (2017 – 2024). First Placement: Tenure-Track Assistant Professor at City University of New York (CUNY), Baruch College, Zicklin Business School.
- Julia Balukonis, USC Marshall (2020 – 2021)

PhD Mentorship — Co-Author:

- Qiran Dong, Berkeley IEOR (2022 – Pres.)
- Yunfan Zhao, Columbia IEOR (2020 – 2023). First Placement: Post-Doctoral Researcher at Harvard Computer Science.
- Brian Rongqing Han, USC Marshall (2016 – 2019). First Placement: Tenure-Track Assistant Professor at University of Illinois at Urbana-Champaign (UIUC), Gies School of Business.

- Michael Hamilton, Columbia IEOR (2016 – 2019). First Placement: Tenure-Track Assistant Professor at the University of Pittsburgh, Katz Graduate School of Business.

Postdoctoral Mentorship — Advisor:

- Ziyu He, Randall R. Kendrick Global Supply Chain Institute (Fall 2023 – Pres.). First Placement: Tenure-Track Assistant Professor, Texas State University.

Qualifying Examination Committee:

- Xinyao Zhang, USC Viterbi ISE (2024); Qing Jin, USC Viterbi ISE (2023); Di Zhang, USC Viterbi ISE (2023); Aikaterini Giannoutsou, USC Marshall OM (2022); Sina Baharlouei, USC Viterbi ISE (2022); Haochen Jia, USC Viterbi ISE (2022); Ying Peng, USC Viterbi ISE (2022); Bo Jones, USC Viterbi ISE (2021); Bradley Rava, USC Marshall Statistics (2019); Shobhit Jain, USC Marshall OM (2018).

Dissertation Committee:

- Xinyao Zhang, USC Viterbi ISE (2026); Di Zhang, USC Viterbi ISE (2024); Ziyu He, USC Viterbi ISE (2023); Ian Yihang Zhu, University of Toronto (2023); Yunfan Zhao, Columbia IEOR (2023); Ying Peng, USC Viterbi ISE (2023); Haochen Jia, USC Viterbi ISE (2022); Bo Jones, USC Viterbi ISE (2022); Brian Rongqing Han, USC Marshall (2020); Michael Hamilton, Columbia IEOR (2019); Junyi Liu, USC ISyE (2019).

Other Mentorship:

- **Shammas Ahmed** (DSO Summer Scholars) Summer 2026
  Project: "Regret Metrics in Decision-Focused Learning." Co-author on a resulting working paper, accepted at the MLxOR Workshop (NeurIPS 2026).
- **Isaac Garee** (DSO Summer Scholars) Summer 2026
  Project: "The Effect of Storage Capacity in Large-Scale Assembly Systems"
- **Yuxuan Ouyang** (USC Master's in Mathematical Finance) Jan. 2025 – Pres.
  Project: "Print a Little, Produce a Lot: 3DP in Large-Scale Assembly Systems"
- **Yongjia Wang** (DSO Summer Scholars) Summer 2024
  Project: "The Fragility of Gerrymandering Metrics"
- **Mengchu Yue** (DSO Summer Scholars) Summer 2024
  Project: "The Fragility of Gerrymandering Metrics"
- **Joshua Zhong** (DSO Summer Scholars) Summer 2024
  Project: "The Fragility of Gerrymandering Metrics"
- **Manan Mehta** (USC Master's in Computer Science) Summer/Fall 2023
  Project: "Enhancing Student Reflections with the Socratic Method using LLMs"
- **Kshitij Ahuja** (USC Master's in Data Science) Summer/Fall 2023
  Project: "Assessing Racial Disparities in Logistics Systems"
- **Arvin Duh** (USC Undergraduate, BUAI) Fall 2023
  Project: Recommending US Colleges
- **Jake Palmieri** (DSO Summer Scholars) Summer 2023
  Project: "Using GPUs to Speed-up Discrete Event Simulation." Winner of an INFORMS Undergraduate Scholarship to attend the annual meeting.
- **Hoang Chu** (DSO Summer Scholars) Summer 2023
  Project: "Assessing Racial Disparities in Logistics Systems." Currently applying to PhD programs in OM / Operations Research.
- **Sara Bangerth** (Former USC Marshall MSBA student) 2022
  Weekly meetings to expose to research and prepare for graduate school.
- **Spencer Xie** (Undergraduate senior thesis) 2021
  "Measuring Digital Upskilling Success." Discovery Scholars distinction; Finalist in the

Discovery Scholars Prize. First Placement: Stanford Law School.

- **Tiffany Chou** (Undergraduate senior thesis) 2021
  "Strategic Use of Patent Portfolios in Cloud Computing Enterprises." Discovery Scholars distinction; 2nd Place USC Libraries Research Prize. First Placement: KPMG.
- **Sanika Sahasrabudhe** (Undergraduate senior thesis) 2021
  "Drivers of Customer Loyalty for Communication Service Providers." Discovery Scholars distinction. First Placement: McKinsey and Co.
- **Xueqi Wang** (UC Berkeley Undergraduate) 2016 – 2017
  First Placement: Duke Biostatistics PhD Program.
- **Qin "Henry" He** (Rising senior, Applied Mathematics/Economics, USC) 2019 – 2020
  First Placement: Duke Statistical Science Master's Program.

University Service

Co-Chair of Undergraduate Curriculum Revision Task Force 2026 – Pres.
Co-chaired a cross-departmental task force reporting to the Dean's Office charged with assessing and revising the undergraduate curriculum. Led benchmarking against peer institutions, reevaluated unit requirements, and developed strategies for meaningful integration of AI into the curriculum.

BUAI Builder Hub (Faculty Co-Lead) 2026 – Pres.
Faculty lead for the BUAI Builder Hub, designed to empower BUAI students to collaborate with faculty on AI tools with positive impact for the university and the broader Los Angeles community. Led the Hub's inaugural projects in partnership with USC Undergraduate Advising and Course Registration.

DSO Summer Scholars Founder, Co-Director 2024 – Pres.
Founded and launched undergraduate summer research program and served as its inaugural director (on sabbatical in 2025). Managed recruitment, personally reviewed all applications, and coordinated participant selection; organized the research-skills workshop series; and secured internal funding through USC Undergraduate Research Associates Program (URAP) and Research Catalyst initiatives.

Data Sciences and Operations (DSO) PhD Coordinator 2021 – 2025
Oversaw the Operations Management and Statistics PhD programs, including program administration, student progress, and coordination across groups.

Data Sciences and Operations (DSO) Peer Evaluation Group 2023 – 2025
Contributed to faculty review processes and supported departmental evaluation. Including 4th Year Review (2023, 2024, 2025).

Operations Management (OM) Group PhD Coordinator 2020 – 2021
Managed the Operations Management PhD program, including admissions, screening and qualifying exams, and ongoing student mentoring.

BUAD 311 Core Course Coordinator 2020, 2021
Coordinated all BUAD 311 instructors to ensure curriculum consistency across sessions, with special focus on unifying course logistics.

Operations Management Group Tenure-Track Hiring Committee (Co-Chair) 2019 – 2020
Led hiring committee, pre-interviewed candidates, and coordinated logistics with faculty and candidates for fly-outs and on-campus interviews.

Data Sciences and Operations – Marketing Seminar Coordinator **2018 – 2019**
Coordinated a bi-semester brown-bag seminar with DSO and Marketing faculty to promote cross-group research collaboration.

Operations Management Group PhD Committee **2017 – 2020**
Helped design curriculum requirements for the PhD program, wrote the optimization screening exam each year, and served on PhD admissions committees.

Data Sciences and Operations Seminar Series Coordinator **2014 – 2019**
Invited visiting faculty to present research in departmental seminar and coordinated all logistical aspects of visits.

BUAD 425 Core Course Coordinator **2017**
Liaised with the Undergraduate Vice-Dean's Office and other core course coordinators to ensure Marshall meets AACSB accreditation standards.

Co-Organizer of 2017 SoCal OR/OM Day **2017**
Organized a one-day, single-track conference with approximately 40 attendees from USC, UCLA Anderson, UC Riverside, and UC Irvine.

Academic Service

Associate Editor — Management Science, Data Science Section **2019 – Pres.**

Associate Editor — MSOM **2021 – Pres.**

Associate Editor — Operations Research, Optimization Section **2024 – Pres.**

ICCOPT 2025 Local Organizing Committee **Fall 2023 – Summer 2025**
Coordinated with the program committee to handle on-the-ground planning for the 2025 ICCOPT Conference (est. 600–1000 attendees).

INFORMS George Nicholson Student Paper Competition **2022, 2023**
Reviewed submissions and assisted in selecting the winner.

INFORMS Junior Faculty Forum Best Paper Award **2022**
Reviewed submissions and assisted in selecting the winner.

INFORMS Minority Issues Forum Student Poster Competition **2021, 2022**
Reviewed submissions and provided detailed feedback to participants.

INFORMS 2021 Annual Conference Organizing Committee **2020 – 2021**
Contributed Sessions Planning Committee: Data-Driven Methods in Optimization.

INFORMS Best Case Competition Judge **2020 – 2021**
Reviewed submissions and assisted in selecting the winner.

Journal Reviewer / Referee:
Operations Research, Management Science, Manufacturing & Service Operations Management, Production and Operations Management, Operations Research Letters, and others.

Conference Program Committee / Reviewer:
NIPS (2016), AAAI (2020), AISTATS (2020, 2023, 2024).

National Science Foundation (NSF) Panel Reviewer:
CMMI / OE Program (2017).